Dr. Rohan Mukherjee

Assistant Professor of Computational Neuroscience

CONTACT

- Bengaluru, India
- r.mukherjee@example.edu
- linkedin.com/in/rohan-mukherjee
- rohanmukherjee.science

AFFILIATIONS

Society for Neuroscience

Bernstein Network

SKILLS & METHODS

Computational modelling

Bayesian inference

Dynamical systems

Large-scale simulation

Python · JAX · PyTorch · Julia · HPC / SLURM

LANGUAGES

English · Bengali · Hindi

MEMBERSHIPS

ACM

Cosyne

Dr. Rohan Mukherjee

Assistant Professor of Computational Neuroscience

Research Statement

My research develops biologically plausible models of learning in spiking neural networks, bridging theoretical neuroscience and machine learning to understand how the brain achieves sample-efficient generalisation.

Education

Ph.D., Computational Neuroscience — 2019

Indian Institute of Science (IISc)

M.Sc., Physics — 2013

IIT Bombay

Appointments

Assistant Professor — 2021 – Present

IIT Delhi — Dept. of Computer Science

Postdoctoral Fellow — 2019 – 2021

University College London — Gatsby Unit

Selected Publications

[1] Mukherjee R., Chen L., Rao S. “Sample-efficient credit assignment in spiking networks.” Nature Machine Intelligence 2024

[2] Mukherjee R., Patel A. “Local plasticity rules approximate backpropagation.” NeurIPS 2023

[3] Mukherjee R., et al. “A dynamical systems view of working memory.” eLife 2022

Grants, Awards & Honors

- SERB Core Research Grant (₹1.2 Cr, 2022–2025)
- Wellcome Trust Early-Career Fellowship (2019)

Teaching

CS6480: Neural Computation (graduate) — 2021–present

IIT Delhi

CS2100: Machine Learning Foundations — 2022–present

IIT Delhi

Presentations

- Invited talk: Plausible deep learning — Cosyne 2024
- Symposium: Neuromorphic RL — NeurIPS Workshop 2023